

Examining functional responses of ant communities to fire in Northwestern African afforested landscapes

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ABSTRACT

In fire prone-regions, forest fires play a vital role in shaping animal communities, particularly for groups like ants strongly associated with vegetation and soil. While prior works on the impact of forest fires on biodiversity focused on taxonomic responses, there is a growing interest on exploring functional responses. This study assesses the taxonomic and functional response of ant species to fire in afforested landscapes (pine plantations) in northern Morocco. These landscapes are widely recognized as suboptimal for many taxonomic groups, suggesting a positive responses of ant communities to fire. Our specific hypotheses were: (1) increasing post-fire openness enhances the richness and abundance of ant species in burnt plots, and (2) fire induces shifts in functional traits within ant communities influencing their composition, potentially favoring species with adaptations suited to the post-fire environment. Our results reveal that fire altered the taxonomic (winner and loser ant species) and functional composition (higher functional richness in burnt plots) of ant communities. Burnt areas harbored a greater abundance of ant species adapted to open habitats, characterized by ground nesting habits, ground foraging, and a preference for readily available resources like seeds. In contrast, unburnt plots hosted species ideally adapted to denser vegetation, utilizing arboreal nesting and foraging strategies and resource niches in the canopy. Overall, we found that fire significantly shapes ant communities by promoting the arrival of species tolerant of open areas and capable of exploiting new available resources in the post-fire habitat.

1. Introduction

Fire is a major disturbance and landscape dynamics driver in terrestrial ecosystems (He et al., 2019). Its influence pervades nearly all regions, profoundly shaping plant composition and structure, along with their associated fauna (Bond et al., 2005). Notably, in ecosystems exposed to fire, wildfire regimes promote resilience of organisms by the acquisition of life-history traits to make them resistant to fire or adapted to early postfire harsh conditions (Keeley and Pausas, 2019). Since forest fires have a major effect on plant and animal communities, it is essential to investigate these responses. This knowledge allows us to predict their impact on biodiversity within forest ecosystems and form a solid basis for management and conservation strategies (Vidal-Cordero et al., 2022).

Until recently, most studies assessing how fire affects biodiversity

tended to focus on changes in taxonomic diversity, measured in terms of richness and/or composition of species (e.g., Basset et al., 2008; Chergui et al., 2020; EL Khayati et al., 2023b). In the past decade however, there is growing recognition of the importance of considering other aspects of diversity, such as functional diversity which has gained prominent as a descriptor of species assemblages (e.g., Arnan et al., 2020; Vidal-Cordero et al., 2023). Functional diversity is assessed by measuring functional traits, which are quantifiable characteristics that have a significant impact on the performance of organisms in their environment (McGill et al., 2006). This metric encompasses the range of ecological processes performed by organisms within a community (Vandewalle et al., 2010). It exhibits sensitivity to how species abundance is weighted, providing insights into community assembly mechanisms (de Bello et al., 2007) and highlighting ecological gradients (Swenson and Weiser, 2010).

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Only by examining fire-related taxonomic and functional responses, we achieve a comprehensive understanding of the ecological consequences of this disturbance (Arnan et al., 2013; Basset et al., 2008). To gain a comprehensive understanding of the mechanisms underlying such responses, it is essential to investigate how abundance of functional traits may change within animal communities (Mouillot et al., 2013). Using this approach, the response patterns of organisms to wildfire can be understood more clearly, and uncover particular adaptations that enhance their ability to survive or recover (Pausas and Parr, 2018).

Ants are highly diverse, abundant, and ecologically important taxa (Hölldobler and Wilson 1990). They exhibit a wide range of morphological and life history traits (Arnan et al., 2014; Parr et al., 2017). They significantly contribute to terrestrial biomass and can modify abiotic and biotic properties of their environment (Retana et al., 2004; Arnan et al., 2011). In particular, ants play a vital role in cycling and aerating the soil, decomposing organic matter, dispersing seeds and protecting plants. (Del Toro et al., 2012).

Ants are very sensitive to a number of disturbances, including prescribed burning, agricultural intensification, and habitat fragmentation (Folgarait, 1998; Underwood and Fisher, 2006). This sensitivity coupled with their ecological importance, makes them valuable subjects in conservation and restoration studies (Campbell and Crist, 2017). Intricate relationship with the environment makes ant communities ideal bioindicator for assessing both taxonomic and functional diversity shifts in degraded forests (Binh et al., 2010; Martello et al., 2018). As key components of forest ecosystems, ants exhibit complex and diverse responses to fire. These responses are influenced by species-specific traits, fire characteristics, and post-fire environmental conditions (Andersen, 1997). Some ant species exhibit remarkable resilience to fire, while others may experience population declines or even local extinctions (Hölldobler and Wilson, 2009).

This study aims to assess how ant species respond to fire in pine plantations located in the African rim of the Western Mediterranean. This region is undergoing a continuous process of degradation as a result of excessive grazing as well as the transformation of natural forest into agricultural (Salah et al., 2018). Afforestation with native and exotic pine species is another land use change occurring in the region (Chergui et al., 2018b). Pine plantations are an important worldwide forest resource that delivers a range of societal, economic and environmental benefits (Cruz and Fernandes, 2008). Along the African rim of the western Mediterranean basin, holm oak (*Quercus ilex*), wild olive (*Olea europaea* var. *sylvestris*), tetraclynis (*Tetraclynis articulata*), and cork oak (*Quercus suber*), are the most widespread native tree species (Benabid, 1982). Yet, in XX century, these native forests were substituted by coniferous plantations, mainly of maritime pine *Pinus pinaster* (Pastor-López et al. 1997), because its wood and resin are valuable socio-economic resources, reflecting the intensity of afforestation (Chergui et al., 2018b). Pine plantations have high (unnatural) tree density, creating a shadier, moister environment that excludes both the natural understory plants and sun-loving, ground-foraging animals, such as ants, who prefer drier conditions (Corley et al., 2006). Dense pine plantations are often associated with a decline in ant diversity (pau-perization) (Andersen, 1995).

Pine plantations currently present numerous ecological challenges, notably the low biodiversity value (da Silva et al., 2019) and its high fire risk due to its high flammability (Keeley et al., 2012).

In Mediterranean regions, the majority of forest fires are intensive crown fires in which habitat openness increases after eliminating vegetation (Keeley and Pausas, 2019). This is particularly important in pine plantations where tree density and canopy are very high (Chergui et al., 2018b). Given that ant communities are responsive to environmental changes (Tiede et al., 2017) and in particular to habitat openness (Andersen, 2019), studying how fire impacts ant communities in these widespread plantations offers valuable insights to inform future decisions towards more ecologically sustainable management practices. Unfortunately, the impact of fire to biodiversity in this region is still not

well studied despite the high level of endemism in many taxonomic groups (Kettani and Moubayed-Breil, 2018).

In this scenario, we first expect that fire will enhance ant species richness and abundance in burnt pine plantations due to increased openness (Anderson and Rodriguez-Cabal, 2019). Canopy destruction by fire can raise soil temperature, potentially restricting the ability of non-thermophilic ant species to use the habitat and favoring heat-tolerant ones (Andersen, 1991; Boulay et al., 2009). Secondly, alteration of plant composition is also likely to impact on food resources availability (Rodrigo and Retana, 2006; Arnan et al., 2007a), modifying the relative abundance of different ant trophic guilds. As result, we expect that burnt areas will harbor ant species whose characteristics are more suitable for open areas (e.g., species that construct ground-based nests and feed on the ground, and species that consume seeds). In contrast, unburnt areas where fire causes the destruction of the nest and the whole colony of vegetation-nesting species (Arnan et al., 2006; Frizzo et al., 2012), will include species adapted to more vegetated habitats (e.g., species that construct nests in vegetation and forage on leaves and nectar). We expect that these functional changes will increase overall functional richness in burnt compared to unburnt afforested plots.

2. Materials and methods

2.1. Study area and sampling design

The study area is situated in the province of Tetouan, in northern Morocco (Fig. 1a). This region features a typical Mediterranean climate with an annual mean temperature of 22 °C (ranging from 8 °C to 35 °C; Ettakifi et al., 2019) and mean annual rainfall of 728 mm (Sellami and Rhinane, 2023), although over the period 2013–2020 the average rainfall has decreased to 452 mm (TuTiempo.net). In the study region, the lithology is limestone ground, and 35 % of the land area is covered by vegetation. Native forests (84 % of woodlands) were historically dominated by Holm oak (*Quercus ilex*), and Thuja (*Tetraclynis articulata*) on limestone substrates (Benabid, 1982). Plantations established between 1942 and 1998 (Houssni et al., 2018) make up the remaining 16 %, with coniferous species like *Pinus pinaster*, *Pinus pinea*, and *Pinus halepensis* comprising the majority (13.04 %) of the planted area (Mharzi Alaoui et al., 2015). Wildfires in this area predominantly take place during the dry season (June–September; Mharzi Alaoui et al., 2015; Chergui et al., 2018a). Moroccan fire cartography reveals that fires in the region tend to be small (typically <1000 ha burnt) compared to other Mediterranean areas due to socio-economic factors such as the density of the rural population and the extensive use of fuels (Chergui et al., 2018a).

We studied ant communities in four sites where pine plantations were impacted by severe crown fires over the period 2012–2016 (Table S1). These sites had a well-established, unburned history of at least 50 years before the fire (Chergui et al., 2020). After the fire, burnt patches were managed with salvage logging (i.e., removal of all the burned wood) (see Khayati et al., 2023b), while at each site, the mid-story vegetation within burnt and unburnt areas consisted of regrowth shrubs including *Arbutus unedo* and *Erica arborea*, and obligate seeders such as *Cistus* sp. Seeders were more abundant in burnt plots, while resprouting species were more prevalent in unburnt plots (EL Khayati et al., 2023a).

Ant sampling was conducted from June through July 2022 using pitfall traps (Agosti and Alonso, 2000; Delabie et al., 2021). A total of 30 plots were established across the four sites, with three sites featuring four burnt plots and four unburnt plots, while the remaining site had three burnt plots and three unburnt plots. At each site, plots were randomly selected and placed at large distances to secure spatial independence within them (Fig. 1b). The average distance among burnt plots (429.14 m ± 54.69 SE), among unburnt plots (489.54 m ± 80.94 SE), and between burnt and unburnt plots (471.29 m ± 67.98 SE) was

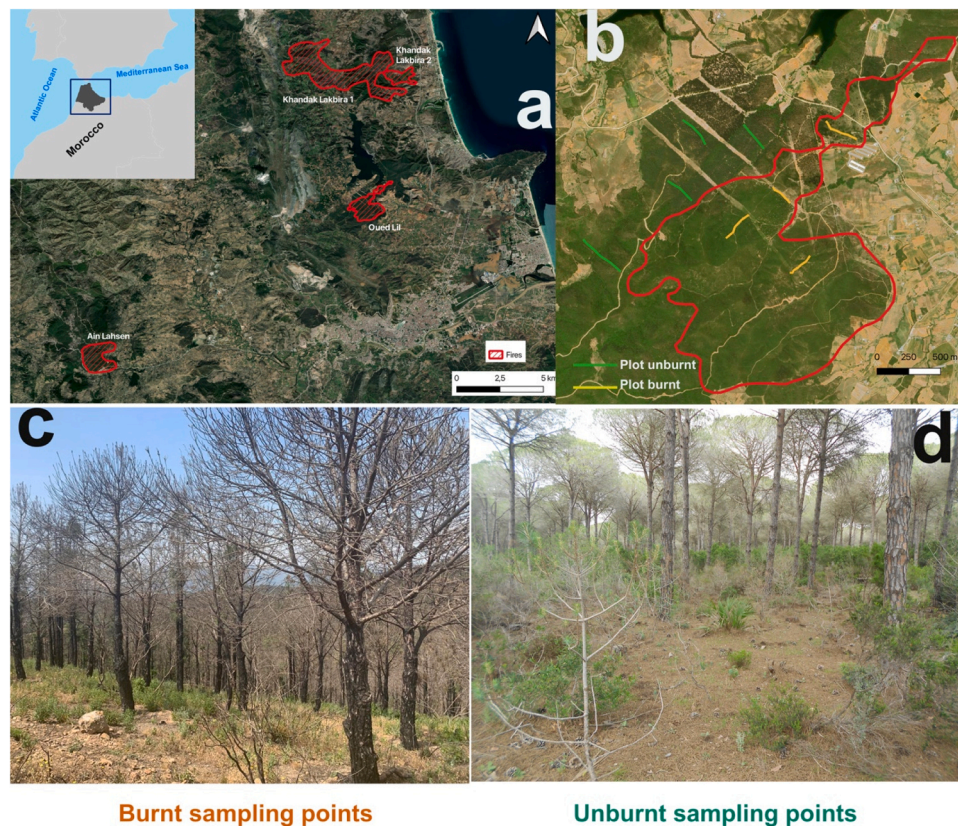


Fig. 1. (a) Location of the four study sites in Tetouan Province, Morocco, with red polygons showing the burnt areas. Location of four burnt and four unburnt plot in the site Oued Lil (b), Pictures of burnt (c) and unburnt (d) plots sampled for ants.

similar. Unburnt and burnt plots were placed in similar pre-fire vegetation to ensure correct comparisons (Fig. 1c and 1d).

At each plot, 6 traps were placed at 10 m intervals along a 50-m linear distance. Overall, there were 90 traps in burnt plots and 90 in unburnt plots. We placed the traps in transparent plastic containers 6 cm in diameter and 8.5 cm high, filled to half their capacity with salt and soapy water. The traps were positioned at least 20 m from the burnt area's edge to avoid collecting ant samples influenced by the transitional zone between burnt and unburnt habitats (ecotone effect) (Vidal-Cordero et al., 2022). During the sampling period, each trap remained active for six consecutive sunny days. On the third day of sampling, we checked the traps and replenished them if they showed signs of evaporating water. Four traps were lost during the sampling and were excluded in the statistical analysis. Following the sampling period, ants were separated from the rest of the invertebrates and stored in 70 % ethanol for later analysis. Ants were identified to species level by the taxonomist author (A. Taheri). Species abundances per plot were calculated by summing ants found at the 6 traps installed per plot. These abundances were used as unit for further analyses.

2.2. Data analysis

2.2.1. Taxonomic analysis

We used Generalized Linear Mixed Models (GLMMs) to compare species richness (number of ant species) between burnt and unburnt plots. We also modelled species abundance (Number of ants detected per plot) for the 13 most abundant species. We used species richness, a widely used measure of biodiversity (Krebs et al., 1999), to assess the response of ant communities to fire. Species richness provides a clear idea of the number of different ant species present in the burned and unburned plots. Although ant abundance data were also collected, they can be influenced by variation in colony size within a species, making

comparisons between plots difficult. For this reason, our primary analysis focused on species richness to provide a reliable measure of community composition.

To normalize data and eliminate heteroscedasticity, the abundance of each ant species was logarithmically transformed prior to analysis (Zar, 1996). Species richness was modelled using a negative binomial distribution to avoid overdispersion, whereas species abundances were modelled using a Gaussian distribution. The fire condition (burnt or unburnt) of plots was used as an independent variable, with site included as a random effect. GLMMs were done with the lme4 package (Bates et al., 2023), and figures with the ggplot2 package (Wickham, 2016).

The similarity of the ant community between unburnt and burnt plots was assessed by permutational multivariate analysis of variance (PERMANOVA) using the "adonis" function in the R package vegan (Oksanen et al., 2022). Pairwise similarity in ant composition among pairs of plots was evaluated using the Bray-Curtis similarity distance for relative abundance data, while the Jaccard similarity index was employed for occurrence (presence/absence) data. We constructed PERMANOVA models using 999 permutations, with fire condition (unburnt or burnt plot) as the main effect, with site as "strata," which limits permutations to within site, and block as a random effect. To visually represent the similarity of transects in terms of species composition, we conducted a non-metric multidimensional scaling (NMDS) analysis. The NMDS has been carried out using the metaMDS function in the vegan package. (Oksanen et al., 2022).

2.2.2. Functional analysis

Each ant species found in pitfall traps was classified in functional classes based on nine life-history traits (Table S2). We selected these traits following Arnan et al. (2013), Retana et al. (2015) and Vidal-Cordero et al. (2022), as they represent three functional trait groups of

acknowledged importance in ant autecology and contribution to survival, persistence and colonization of burnt landscapes. We first constructed a matrix of trait abundances recorded at each plot, and these abundances were modeled using Generalized Linear Mixed Models (GLMMs), with fire condition (unburnt or burnt) serving as the independent variable. Abundances were log transformed to prevent data overdispersion, and GLMMs performed with a Gaussian distribution. Site was incorporated as a random effect to account for potential site-specific disparities and to ensure independence. The GLMM was performed using the lmer function in the lme4 package (Bates et al., 2023).

Trait abundances per plot were used to compute three primary functional diversity (FD) indices, following the approach outlined by Mason et al. (2005): functional richness (FRic), functional dispersion (FDis), and functional evenness (FEve). Functional richness is a measure of the volume of the convex hull of the functional space that the community occupies (Villéger et al., 2008). Functional evenness assesses the uniformity of distribution of abundance in the space of functional traits (Mason et al., 2005). Functional dispersion is defined as the mean distance in the multidimensional trait space of individual species to the centroid of all species (Laliberté and Legendre, 2010). These analyses were conducted using the dbFD function of the FD package (Laliberté et al., 2023). We assessed Differences in FD indices between burnt and unburnt plots were examined with GLMMs adjusting data to a Gaussian distribution, with site as a random effect.

3. Results

3.1. Taxonomic results

Across all plots, we collected a total of 9073 ant workers belonging to 45 species and 20 genera (Table S3). Of these, 11 species were exclusive from unburnt plots, 11 from burnt plots and specimens of 23 species were found both in unburnt and burnt plots (Figure S4). We did not find differences in species richness between unburnt and burnt plots (Table 1). The five most abundant species in the plots were *Monomorium salomonis* (47 % of abundance), *Camponotus cruentatus* (12 %), *Pheidole pallidula* (9 %), *Camponotus ali* (7 %) and *Crematogaster auberti* (5 %). GLMMs for the 13 most abundant ant species showed opposing fire responses: *M. salomonis* and *C. auberti* were more abundant in burnt plots,

Table 1

Results of generalized linear models with gaussian error structure of the abundance of the 13 most abundant ant species in unburnt and burnt plots in north-western Africa. Significant p values in bold.

Species	Number of individuals		Fire condition		
	Unburnt	Burnt	Estimate	Std. Error	P
Species richness	35	34	-0.112	0.108	0.304
<i>Monomorium salomonis</i>	580	3682	-0.524	0.273	0.055
<i>Camponotus cruentatus</i>	826	289	0.561	0.159	0.0004
<i>Pheidole pallidula</i>	425	363	-0.156	0.244	0.523
<i>Crematogaster auberti</i>	105	324	-0.449	0.214	0.0361
<i>Camponotus ali</i>	288	369	0.021	0.180	0.906
<i>Aphaenogaster gemella</i>	119	118	0.313	0.241	0.090
<i>Aphaenogaster mauritanica</i>	102	64	0.015	0.210	0.943
<i>Aphaenogaster senilis</i>	101	110	0.005	0.205	0.981
<i>Plagiolepis maura</i>	105	154	0.009	0.190	0.960
<i>Tetramorium caespitum</i>	64	162	-0.330	0.205	0.108
<i>Temnothorax curtulus</i>	61	5	0.683	0.161	0.0003
<i>Camponotus serotinus</i>	51	14	0.120	0.140	0.393
<i>Plagiolepis schmitzii</i>	23	62	-0.156	0.144	0.278

Additionally, some species were exclusive of one fire-type plot: *Crematogaster scutellaris*, *Temnothorax spinosus* and *Camponotus lateralis* were only found in unburnt area, whereas *Messor sanctus*, *Messor barbarus* and *Tetramorium semilaeve* were only found in unburnt area (Table S3).

C. cruentatus and *T. curtulus* in unburnt plots, whereas the remaining most abundant species showed no significant differences between unburnt and burnt plots (Table 1).

In accordance with the GLMMs, the PERMANOVA results uncovered significant differences in both species abundance (Bray-Curtis's index) ($F=2.4202$; $R^2=0.0795$; $p=0.005$) and species occurrence (Jaccard index) ($F=1.9074$; $R^2=0.0638$; $p=0.007$) between unburnt and burnt plots. The NMDS showed taxonomic differences in ant communities with axis 1 discriminating between unburnt and burnt plots, with a stress value of 0.2154 (Figure S5).

3.2. Functional results

Of the 25 ant functional classes, nine showed positive responses to fire (Table 2), including hypogaeic-nesting species, species that are strictly diurnal, species with medium worker polymorphism, polygynous species, species that display dependent colony foundation, species with small body size, species that use individual and collective foraging, and species that consumed seed-based diet. In contrast, four functional classes showed negative responses, including species with larger body size, species with greater worker polymorphism, species that display independent colony foundation, and arboreal nesting species (Table 2; Fig. 2).

Overall, functional richness increased significantly from unburnt to burnt plots, functional evenness decreased from unburnt to burnt plots (Table 3; Fig. 3), and functional dispersion remained unchanged with fire condition.

4. Discussion

Our field study conducted in pine plantations from northwestern Africa demonstrates that fire can alter the short-term taxonomical and

Table 2

GLMM results of functional trait abundances of ant communities sampled at 4 sites of north-western Africa, according to fire condition (unburnt, burnt). Negative estimates correspond to functional classes more abundant in burnt plots. Significant p values in bold.

Functional trait	Strates	Fire condition		
		Estimate	Std. Error	P
Worker body size	Small <ou=4	-0.210	0.065	0.0012
	Medium 4<X<ou=8	-0.032	0.058	0.644
	Large >8	0.603	0.135	< 0.0001
Worker polymorphism	Low <ou=0.3	-0.051	0.140	0.709
	Medium 0.3<X<ou=0.6	-0.290	0.096	0.0024
	High >0.6	0.095	0.043	0.0278
Colony size	<ou= 500	0.043	0.145	0.759
	500<X<ou=1000	-0.030	0.158	0.849
	>1000	-0.082	0.057	0.225
Diet	Seed	-0.187	0.063	0.0028
	Insects	-0.056	0.048	0.334
	Liquid	0.063	0.049	0.256
Colony foundation	Dependent	-0.258	0.074	0.0005
	Independent	0.175	0.075	0.0189
	Both DCF and ICF	-0.068	0.255	0.787
Number of queens	Monogyny	0.002	0.046	0.889
	Polygyny	-0.656	0.211	0.0019
	Both monogyny and polygyny	-0.114	0.245	0.635
Diurnality	Non strictly diurnal	-0.267	0.170	0.109
	Strictly diurnal	-0.141	0.066	0.0321
Foraging strategy	Individual	-0.366	0.192	0.0565
	Group	-0.001	0.061	0.324
	Collective	-0.558	0.254	0.0279
Nest site	arboreal	0.334	0.112	0.0028
	hypogaeic	-0.090	0.039	0.0188

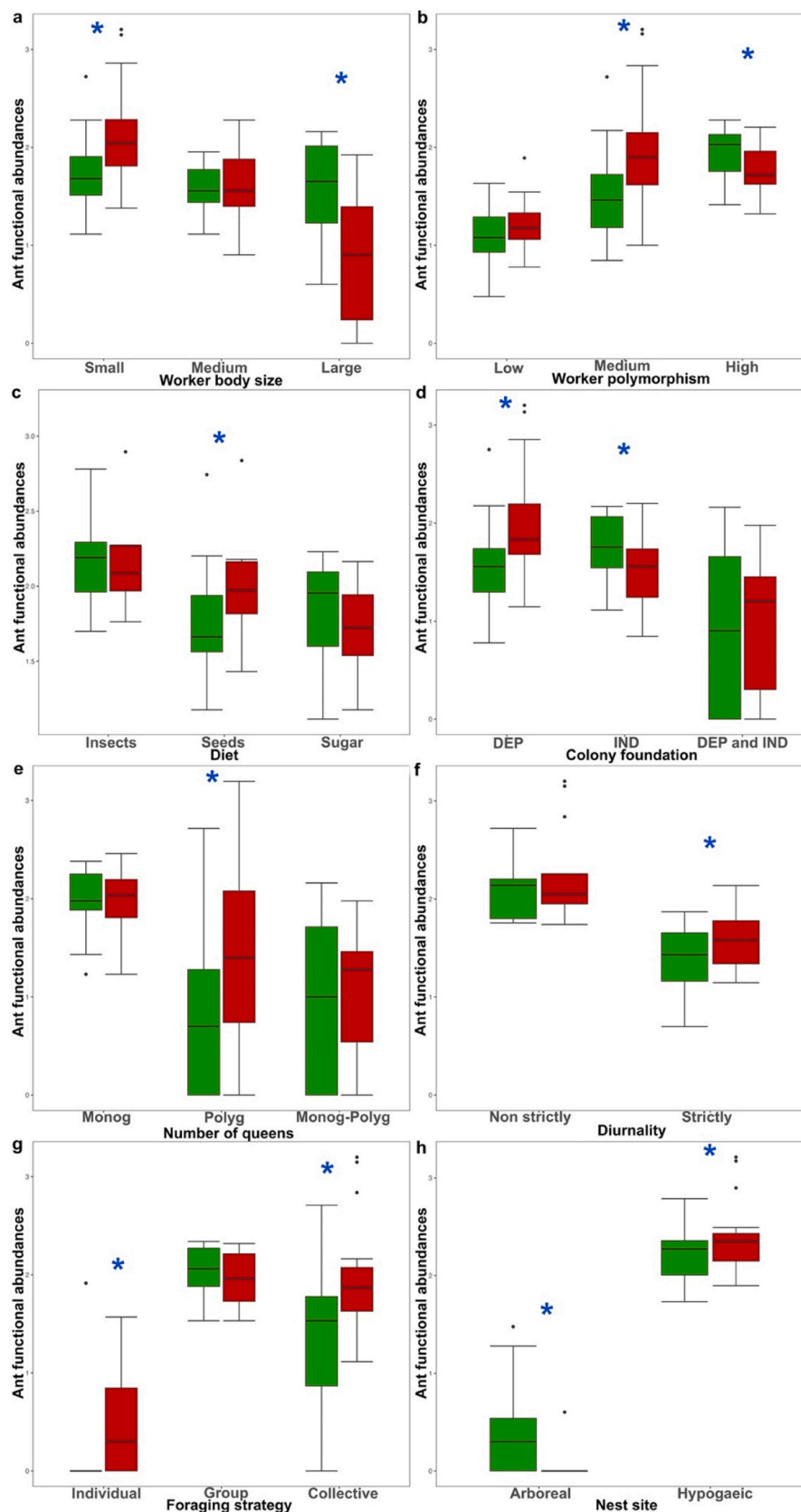


Fig. 2. Differences in the abundance of ant functional classes between unburnt (green) and burnt (red) plots in four pine plantations from northwestern Africa. Mean (lines), boxes (standard errors) and whiskers (standard deviation) are obtained from Generalized Linear Mixed Models. Stars show significant variations in terms of abundance between unburnt and burnt plots.

Table 3

Results of the analysis using GLMMs for three measures of the functional diversity of the ant community at 4 sites of north-western Africa, according to fire condition (unburnt, burnt). Significant p values in bold.

Functional index	Fire condition		
	Estimate	Std. Error	P
Functional richness (FRic)	-0.189	0.081	0.0201
Functional evenness (FEve)	0.029	0.013	0.0219
Functional dispersion (FDis)	0.004	0.125	0.973

functional composition of ant communities.

At the taxonomic level, we detected a shift in the community composition with winner and loser ant species caused by fire. This finding aligns with previous studies that have documented the effect of fire on ant communities in other regions (e.g., Parr et al., 2004; Andersen, 2000a; Sondej and Domisch, 2022; Arnan et al., 2013). Notably, at the functional level, we detected a significant shift in dominant functional trait classes. Overall, there was greater ant functional richness in burnt than in unburnt plots. The emergence of a new, open postfire habitat favoring thermophilic ant species with specific adaptations (e.g., ground nesting, heat tolerance) may be the underlying cause of this functional trend.

This positive functional postfire response is context-dependent and can vary depending on the habitat studied (e.g., native forests and pine plantations). Dense pine plantations have low ant abundance and species richness, a trend reported in many regions such as Australia (Sinclair and New, 2004), South Africa (Samways et al., 1996), and Argentina (Corley et al., 2006; Santoandré et al., 2019), countries where pine is a widely cultivated exotic forest species as the Mediterranean basin. In Morocco, this trend has also been observed with lower ant diversity in afforested than native ecosystems (Fernane et al., 2023). We acknowledge that we lack a comparison in ant responses to fire between native forests and pine plantations. However, the general ant community trend observed in pine plantations is indicative that fire can improve the habitat for some ant functional traits. Since pine plantations are so prevalent around the world, our results may provide useful insights into post-fire management in these habitats.

4.1. Taxonomic responses of ants to fire

We observed opposing responses to fire within the commonest ant species. For example, *M. salomonis* and *C. auberti* were more abundant in burnt plots. These species are typically found in the warmer areas of the Mediterranean region and exhibit ground-nesting behavior with dense populations (Cagniant, 2005, 2006). Post-fire environments are characterized by open habitats and wide availability of food resources (Arnan et al., 2019), which were favorable for the postfire establishment of these species. Top of Form Conversely, fire poses a significant challenge for ant species like the forest-dwelling *C. cruentatus* and the more cryptic *T. curtulus*. These species, heavily reliant on vegetation for nesting and foraging, are unable to adapt to the post-fire environment due to the lack of suitable microhabitats (Arnan et al., 2006). Consequently, they were exclusively found in unburnt pine plantations within the study sites.

The remaining most abundant ant species (*Pheidole pallidula*, *Aphaenogaster senilis*, *Tetramorium caespitum* and *Plagiolepis schmitzii*) showed no significant differences between unburnt and burnt plots, suggesting that these species have a high ecological valence/flexibility which can adapt to a wide variety of habitats and micro-environments. Regarding their life-history traits, these species show remarkable flexibility in their nesting preferences, since they readily utilize diverse structures, including the ground, dead spindles, beneath bark, lodges, and galleries beneath stones (Gomy et al., 2022). This adaptability enables them to evade the direct impact of fire, such as heat, drought, and smoke (Antunes et al., 2009). The occurrence of a high number of generalist species within a community reduces differences between unburnt and burnt plots (Santos et al., 2014). However, the low number of specialist ant species can also reflect the poor quality of pine plantations for ant communities (Mateos et al., 2011), as occurs for other groups of terrestrial invertebrates (Da Silva et al., 2019).

4.2. Functional responses of ants to fire

One particularly striking finding is the significant increase in functional richness observed in burnt plots. This may be due to the fact that

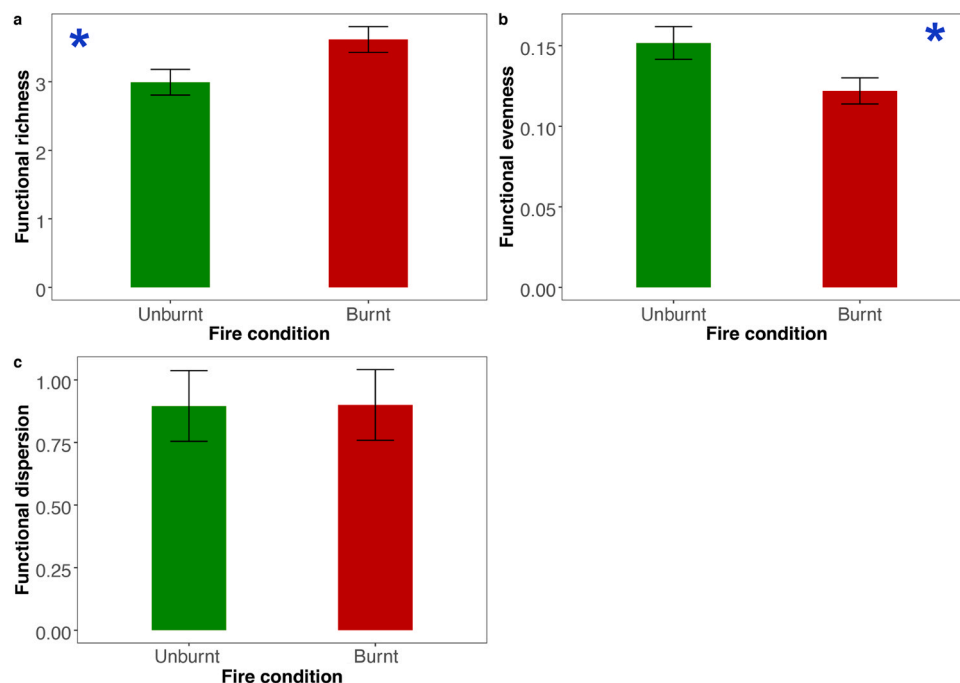


Fig. 3. Differences in ant functional richness (a), functional evenness (b) and functional dispersion (c) between unburnt (green) and burnt (red) plots. Bars are mean and whiskers standard error values of Generalized Linear Mixed Models estimates.

burned plots are more diverse in terms of environmental niches, offering opportunities for a wider range of ant species (Kosickia and Hromada, 2018) compared to dense pine plantations. A wider diversity of foraging strategies (individual and collective foraging) in burnt areas could be linked to the greater volume and occupancy of niches in burnt plots compared with unburnt plots (Arnan et al., 2013).

In pine plantations, fire structurally alters the habitat by clearing existing vegetation and generating open spaces, conditions readily exploited by fire-adapted ant species (Arnan et al., 2006; Rodrigo and Retana, 2006). These newly available niches allow for the coexistence of a more diverse range of ant species with different functional traits, contributing to the observed increase in functional richness. This pattern can be explained by different mechanisms:

Firstly, the changes induced by fire in pine plantations are particularly significant, as these high canopy forests typically have a sparse and depauperate understory due to the acidification caused by fallen pine needles (Peltier et al., 2001). Therefore, fire acts as a reset button, clearing the acidic layer and stimulating plant growth, thereby enhancing understory plant richness and habitat heterogeneity. This, in turn, enhances the availability of resources for arboreal ant species, such as microhabitats for nesting and favorable abiotic conditions for foraging (Joern and Laws, 2013).

Overall, the impact of fire on vegetation can promote herbaceous and shrub species to regrow and germinate (Podgaiski et al., 2013; Lamont and Downes, 2011). These changes in vegetation can attract certain ant species that may benefit from the post-fire increased abundance of food (Moretti et al., 2002; Schafers et al., 2008). We acknowledge that the effect of fire on ant assemblages can greatly vary among ecosystems. While the effect of fire on ant diversity was not significant in deserts, grasslands and savannas, it was negative in forests, especially tropical forests (Vasconcelos et al., 2017; Kuchenbecker et al., 2023). Such findings strengthen the idea that the effects of fire on ant assemblages depend on the vegetation complexity and the magnitude of fire-associated changes in vegetation structure (Sackmann and Farji-Brener, 2006; Philpott et al. 2010). In pine plantations, the increased plant richness and diversity following a fire (EL Khayati et al., 2023a) can provide ants with a broader range of resources, including nectar, pollen, seeds, and insects. Additionally, the flowering of plants after a fire can attract pollinators, providing a food source for ants that specialize in preying on insects (Lamont and Downes, 2011). This increased food availability can support a larger population of ants and attract new functional groups to burnt areas.

Secondly, fire affects ant communities by creating new habitats and niches. In areas with high water deficits or a high percentage of bare soil, such as many parts of the Mediterranean, ant communities contain more species adapted to burned areas (Arnan et al., 2006). This is because fire creates conditions similar to drought-prone open environments with limited vegetation cover, offering new opportunities for ant species. Additionally, water deficit can intensify interspecific competition, favoring species that thrive in open areas and tolerate higher temperatures (Cerdà et al., 1997; Cerdà et al., 1998). The plant species present in the burned area, particularly in areas dominated by *P. pinaster* and *P. halepensis*, play a crucial role in shaping ant communities. These provide a variety of resources and microhabitat for different ant species; pine forests can gradually reestablish after fire. These pines produce serotinous cones, which require heat from fire to open and release their seeds, serving as valuable food resources for ants (González-De Vega et al., 2016; Elvira et al., 2021; Madrigal et al., 2021). This phenomenon could partially explain the higher ant functional richness observed in burnt pine plantations compared to unburnt ones in our study.

Another remarkable finding of our study showcases a distinct shift in functional composition between burnt and unburnt plots, aligning with our hypothesis that fire reshapes ant communities by favoring specific trait classes. Burnt areas harbored a greater abundance of ant species equipped for open habitats, characterized by ground-nesting habits, ground foraging, and a preference for readily available resources like

seeds. Conversely, unburnt plots hosted species optimally adapted to denser vegetation, utilizing arboreal nesting and foraging strategies alongside niche resources within the canopy. This divergence in functional traits supports the findings of Maeto and Sato (2004), who observed increased abundance of open habitat specialists in disturbed forests, mirroring the trend we observed in burnt areas.

Functional shifts caused by fire can be explained by several reasons:

- Firstly, the chances of surviving the fire were boosted by several functional traits. Soil-nesting or burrow-digging ants demonstrate a greater rate of survival compared with species that live tightly associated with vegetation (Matsuda et al., 2011). Ant species nesting within the vegetation directly succumb to the effects of fire. However, the majority of ground-nesting species outlive fire as there is hardly any difference in temperature a few centimeters beneath the surface of the ground. (Arnan et al., 2006; Frizzo et al., 2012). The removal of vegetation cover through fire may raise the temperature of the ground and restrict habitat utilization by non-thermophilic ant species (Andersen, 1990, 1991, Boulay et al., 2009), resulting in selection for heat-tolerant traits (Arnan et al., 2013).

- Secondly, post-fire persistence was favored by particular foraging traits. Studies have previously demonstrated that the area on which ants feed (e.g., vegetation or soil) as well as feeding resources exploited may affect their persistence (Arnan et al., 2007a; Rodrigo and Retana, 2006). Our study revealed that vegetation-feeding ant species were more abundant in unburnt plots, while burnt plots attracted species with seed-based diet and those utilizing both individual and collective foraging strategies. This aligns with the radical changes in food resource availability following a fire (Arnan et al., 2007a). Burnt areas become more open, creating conditions favorable for an increase in seed abundance from herbaceous successional plants. This aligns with our previous study in which obligate seeders appeared to be more abundant than resprouters in afforested plots (Vidal-Cordero et al., 2022; EL Khayati et al., 2023a), which may explain the finding of ants that use a collective foraging strategy, as this strategy prevails in more open areas (Arnan et al., 2014). Collective foraging strategy enables the recruitment of many workers for concentrated sources of food (Cerdà et al., 1997; Arnan et al., 2012). As the removal of vegetation cover by fire can increase soil temperatures, species foraging when temperatures are warmer (i.e., midday) are individual foragers that do not rely on chemical signals for nestmate recruitment (Ruano et al., 2000), presumably since elevated soil temperatures restrict the efficiency of these signals. (van Oudenhove et al., 2012; Retana et al., 2015). Additionally, the increased thermal contrasts and reduced vegetation cover in burned areas may favor the establishment of strictly diurnal ant species, as they can forage more effectively in these open habitats without the risk of overheating or predation (Nathan and Ne'eman, 2000; Arnan et al., 2013).

Moreover, granivory becomes particularly advantageous in arid and semi-arid regions due to the ability to store dry seeds for extended periods, bridging gaps in food availability during dry seasons (Buckley, 1982; Rico-Gray and Oliveira 2008). This aligns with our previous findings, where obligate seeders exhibited better performance compared to resprouters in the short term within highly disturbed and dry pine plantations (EL Khayati et al., 2023a). Interestingly, our findings diverge from those of Arnan et al. (2019) who suggested a preference for sugars and/or insects among Mediterranean ants. This discrepancy underlines the potential context-dependence of ant dietary choices, highlighting the crucial role of habitat characteristics and resource availability in shaping their foraging strategies.

- Thirdly, post-fire conditions appeared to favor the persistence of species with medium worker polymorphism in burnt plots, while unburnt plots hosted species with high worker polymorphism. Worker polymorphism, the variation in worker size and morphology within an ant colony, has been linked to the ability of colonies to adapt to environmental challenges (Mertl and Traniello, 2009; Arnan et al., 2011). Studies suggest that colonies with higher polymorphism exhibit greater

variability in temperature responses, increased total external activity, and improved foraging efficiency, leading to better adaptation (Cerdá and Retana, 1997). However, our study found surprisingly higher worker polymorphism in unburnt plots compared to burnt plots. This divergence could be due to specific predator pressure in unburned areas, favoring different castes of workers for defense or foraging, or resource limitation in burned areas, selecting for less specialized workers able to exploit a wider range of resources due to niche overlap. Additionally, our results suggest that small workers may be more prevalent in burned plots due to their ability to tolerate higher temperatures and forage more effectively in open habitats (Bishop et al., 2021).

- Fourthly, the post-fire colonization of burnt plot was facilitated by dispersal traits of ant species. Ant species recorded in burned plots primarily utilize dependent colony foundation, a dispersal strategy where the young queen is assisted by workers, a mode generally found in polygynous ant species (Passera and Aron, 2005). In contrast, unburnt plots hosted ant species that utilize independent colony foundation, a dispersal strategy where a freshly fertilized queen, feeding her first larvae herself, is at the origin of a new society (Passera and Aron, 2005). A possible explanation for this is that species in burnt plots utilize dependent colony foundation due to the post fire openness that makes these sites beneficial for predators due to the increase in insect food supply (Engstrom, 2010). In species that use dependent colony foundation, foundation is secured compared to species that use independent colony foundation (Armor et al., 2011). Indeed, even if the foundation of independent colonies provides dispersal advantages at long distances (Armor et al., 2011), it presents considerable weaknesses for new colonies confronted with predation and competition (Wiernasz and Cole, 2003; Boulay et al., 2007).

4.3. Concluding and conservation remarks

Ants play a key role in ecosystem functioning globally (del Toro et al., 2012), with fire significantly affecting terrestrial arthropod communities, leading to changes in taxonomic composition and short-term functional shifts. Our study in pine plantations revealed a clear shift in ant community composition following fire, highlighting fire-influenced winners and losers, as well as changes in dominant functional trait classes. Despite the economic benefits of pine plantations, their low biodiversity and ecological homogeneity make them unsuitable for different animal groups, including arthropods (da Silva et al., 2019). Our results highlight the unsuitability of pine plantations as habitats for Mediterranean ant communities, especially due to the prevalence of generalist species. This study improves our understanding of ant responses to fire disturbance and sheds light on traits important for survival, persistence and colonization.

Assessing arthropod responses following disturbance can serve as early indicators of ecosystem change (Kremen et al., 1993; Lawes et al., 2005), providing insights for management and conservation planning, particularly in areas where forestry and biodiversity conservation overlap. While our study focused on short-term intervals, we advocate longer-term investigations to deepen our understanding. Future research should explore the responses of ant communities to fire in northwestern African native forests, the effect of additional fire characteristics, including fire intervals and severity, as well as alternative options for post-fire management of burnt areas (Vidal-Cordero et al., 2023). These efforts will significantly enhance the capacity to forecast responses of ecosystems, assess biodiversity and design conservation strategies to address current global change (Vidal-Cordero et al., 2022).

CRediT authorship contribution statement

Ahmed Taheri: Writing – review & editing, Validation, Methodology, Investigation. **Soumia Fahd:** Writing – review & editing, Visualization, Supervision, Investigation. **Xavier Santos:** Writing – review & editing, Visualization, Supervision, Methodology, Investigation,

Conceptualization. **Brahim Chergui:** Writing – review & editing, Supervision, Methodology, Investigation, Formal analysis. **Mounia EL KHAYATI:** Writing – original draft, Methodology, Investigation, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have influenced the work reported in this paper.

Data Availability

Data will be made available on request.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.foreco.2024.122023.

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